CS422 Computer Networks (Fall 2025): Homework 3

(Due Date: 23:59:59PM Thursday Oct 23, 2025, Total: 40 points)

1. **(5 points)** TCP waits until it has received three duplicate ACKs before performing a fast retransmit. Why do you think the TCP designers chose not to perform a fast retransmit after the first duplicate ACK for a segment is received?

2. **(5 points)** Why is a 2-way handshake not sufficient for reliable TCP connection establishment, and why do we need a 3-way handshake instead?

3. **(6 points)** After one party sends the final ACK in response to the peer's FIN segment during TCP connection termination: Why must this host remain in the waiting state for a certain period before fully closing the connection? If another FIN segment arrives before the timer expires, what will happen, and how does the host handle it?

4. **(6 points)** What is the difference between flow control and congestion control in TCP? Assume the CWND is 10 MSS, while the RWND is 8 MSS, what will be the effective transmission window size?

5. **(6 points)** TCP congestion control uses AIMD for fairness. Please think about what will happen if TCP CC uses AIAD (additive increase and additive decrease (increase/decrease by one)) or MIAD (multiplicative increase and additive decrease). Please follow the diagram plotted in the lecture to see what will happen with two flows.

6. **(12 points)** Assume TCP congestion control uses all the following algorithms covered in our lecture: slow start (SS), congestion avoidance (CA), fast retransmit and fast recovery, and retransmission upon timeout. All the following assumptions are the same as those used in class unless specified (say, cwnd, ssthresh and which packet gets lost).

   - MSS = 1B. Initially *ssthresh* at the sender is set to 4, and *cwnd* as 1. Assume *cwnd* and *ssthresh* are measured in segments (or in bytes as MSS = 1B). The transmission time for each segment is negligible. Retransmission timeout (RTO) is initially set to 500 ms at the sender and is unchanged during the connection lifetime. RTT is 100ms for all transmissions.
   - The connection starts to transmit data at time t = 0, and the initial sequence number starts from 1. Segment with sequence number 9 is lost once. No other segments are lost.
   - The receiver acknowledges every segment (no delayed ACK) and uses TCP cumulative ACK (i.e., ACK = (largest sequence received in order at the receiver + 1)). Besides, the sender always has data available for transmission.
   - When *ssthresh* equals to *cwnd*, we use slow start, not congestion avoidance.
   - Right after fast retransmit/fast recovery phase, if *ssthresh* equals to *cwnd*, use the slow start algorithm.
   - The receiver will buffer out of order packets (like selective repeat).

   When does the sender receive the ACK for the segment with the sequence number 15? Which algorithms will be used upon receiving ACK for the segments with the sequence number 15 and 16? Show your diagram. (hints: please follow the example diagram in class to show congestion control step-by-step.)